Fiscal Year:	FY 2018	**Task Last Updated:**	FY 08/09/2018
PI Name:	Khusid, Boris Ph.D.		
Project Title:	Kinetics of Electric Field-Driven Phase Transitions in Polarized Colloids		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	khusid@njit.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	973-596-3316
Organization Name:	New Jersey Institute of Technology		
PI Address 1:	Chemical & Materials Engineering		
PI Address 2:	University Heights		
PI Web Page:			
City:	Newark	**State:**	NJ
Zip Code:	07102-1982	**Congressional District:**	10
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2013 Complex Fluids & Macromolecular Biophysics NNH13ZTT001N
Start Date:	08/23/2013	**End Date:**	08/22/2019
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Duval, W.	**Contact Phone:**	216-433-5023
Contact Email:	walter.m.duval@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 8/22/2019 per NSSC information (Ed., 8/7/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Nelson, Emily Ph.D. (NASA Glenn Research Center) Sheehan, Christopher M.S. (ZIN Technologies, Inc.) Loewen, Hartmut Ph.D. (University of Dusseldorf) Telo da Gama, Margarida Ph.D. (University of Lisbon) Yethiraj, Anand Ph.D. (Memorial University of Newfoundland)		
Grant/Contract No.:	NNX13AQ53G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Motivation: Colloidal assembly provides the capability for scalable manufacturing of structured materials and devices for traditional and emerging industries. Although equilibrium structures and thermodynamics of colloids exposed to a field have been well studied, the use of electrically assisted assembly is limited by a lack of understanding of the role of time and spatially varying fields in directing and controlling non-equilibrium processes. Microgravity offers an opportunity to study field-driven phenomena on non-buoyancy-matched colloids by removing masking gravity effects, such as particle sedimentation, convection, and jamming. Objectives: Design and conduct microgravity tests in the International Space Station Advanced Colloids Experiment (ISS ACE-E) facility in long timescales to elucidate fundamental microscopic mechanisms of non-equilibrium phenomena underlying the field-assisted assembly of interacting colloidal particles and suggest novel routes for creating functional materials. The emphasis will be on the field-driven evolution of the mesoscale phase transitions, electro-hydrodynamic instabilities, and nucleation and growth of crystalline structures. The phenomena selected are logical candidates to take advantage of microgravity because terrestrial observations have failed to provide a comprehensive physical understanding of the underlying physics. Methodology: Dielectrophoretic experiments will be carried out on well-characterized colloids under well-defined conditions in terrestrial and microgravity environments. We use electro-hydrodynamic methods and apparatuses for the control and manipulation of particles which were developed in our NASA-supported projects, including parabolic flight tests conducted in 2002 and 2013. Molecular dynamics simulations, coarse-grained and continuum models will be employed to study field-induced colloidal structures. Deliverables: Detailed understanding of electrically driven non-equilibrium phenomena in colloids, apparatuses for the field-assisted colloid assembly, experimental database, and predictive models for the field-induced structure. The research will be integrated with educational and outreach initiatives in space sciences and related technologies.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research has the strong potential to reveal the fundamental microscopic mechanisms of electrically driven phenomena in a suspension of interacting particles and develop a physical explanation for puzzling observations under terrestrial conditions that are not predicted by current theories. The main focus is on the appearance of a cellular structure in a colloid exposed to a spatially uniform field and understanding of the nucleation and growth of crystalline structures in a high-gradient and mechanisms of electro-hydrodynamic instabilities of concentration fronts. The obtained results will facilitate the development of new technologies for the control and manipulation of colloidal processes. Space applications: Electrical technologies can be employed in space to mitigate the effects of reduced gravity by providing an electrical body force. Results of ACE-T-E experiments will be useful for the development of new methods for electro-hydrodynamic control, manipulation, and separation of multiphase fluid flows in various space applications, for example, thermal management systems. Earth applications: The research should bring the understanding of electrically driven colloidal processes to the level useful in the design of apparatuses and processes for a broad range of applications in materials engineering, life science, electronics, and chemical industry. This will stimulate further "spin-offs" of electrical technologies.
Task Progress:	Major Accomplishments: The study was conducted within the scope of the originally-proposed research plan. To specify basic parameters for the design of the ACE-E cell geometry, electrode configuration, voltage/frequency ranges, and procedures of preparing suspensions for the ISS experiments, the New Jersey Institute of Technology (NJIT) team carried out ground-based experiments on suspensions of nearly neutrally buoyant polyethylene (53-75 μm and 40-48 μm) and polyalphaolefin (4-5 μm) particles dispersed in several low conducting solvents (Mazola corn oil, Envirotemp FR3 and E200 Cargill dielectric fluids and silicone oil with non-ionic surfactant Brij 30). It was demonstrated that cellular structures are formed by exposing a suspension to a combination of a strong AC field and a weak DC field. The range of parameters needed to form these structures is mainly determined by dipolar forces between the particles. While both AC and DC fields are required, the processes of forming cellular structures depends on the way how these fields are applied. Presented results summarize a sequence of 3D->3D processes following the application of an AC field and then adding a DC field and a sequence of 2D->3D processes following the application of a DC field and then adding and AC field. Development of Requirements for ISS experiments: The NJIT team worked in closed collaboration with NASA scientists on the design of hardware for ISS ACE-E experiments. The main focus was on the development of requirements for a high-voltage power supply and an optical system that are summarized in ACE-E Experiment Summary. The Principal Investigator (PI) presented the main results of this work at the Science Concept Review meeting at NASA Glenn Research Center on November 8, 2017. The Science Review Committee stated that the NJIT team had addressed their concerns and potential problems expressed at the Science Concept Review meeting on November 10, 2015. They stated that the output voltage and frequency ranges of the function generator and amplifier needed for the ISS experiments were consistent with other ACE-E experiments and with commercially available off-the-shelf (COTS) equipment. Requirements for an optical system in the ISS experiments were demonstrated to be consistent with the current ISS Light Microscopy Module (LMM) system. The materials for proposed experiments were fully characterized. The Committee also confirmed the adequacy of the proposed test matrix and mission success criteria and stated that the ground-based testing had been taken about as far as possible. Need for No-Cost Extension and Planned Work for Next Year: The NJIT team need a one-year no cost extension to complete the following scheduled tasks: 1. Continue working with NASA project scientists and ZIN Technologies personnel on the selection of a high-voltage amplifier, voltage/frequency regimes, ACE-E cell design, and suspension compositions for ISS tests. 2. Continue working of the development of a theory for formation of cellular structures. 3. Continue ground-based experiments needed to finalize science requirements for experiments aboard the ISS.
Bibliography Type:	Description: (Last Updated: 02/06/2025)

Abstracts for Journals and Proceedings	Khusid B. "Requirements for ISS ACE-E Cell Design to Study Electric Field-Driven Structuring in Suspensions." Presented at 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. Concurrent Session: Complex Fluids. 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. , Oct-2017
Abstracts for Journals and Proceedings	Khusid B, Elele E, Lei Q, Tang J, Shen Y. "Single-bubble boiling under Earth's and low gravity." Presented at the 70th Annual Meeting of the APS Division of Fluid Dynamics, Denver, CO, November 19-21, 2017. Bulletin of the American Physical Society. 2017 Nov;62(14):BAPS.2017.DFD.A7.8. http://meetings.aps.org/link/BAPS.2017.DFD.A7.8 , Nov-2017